

The Supply Edge: A Mass-Conserving Rule Without Tuned Radii Fixes Structure Growth Under a Vacuum-Tied MOND Law

Working note

The authors — 2026-10-08, version 2.0 (v1.0, 2026-10-07, stated the hypothesis; this version adds the simulations that test it) — AI-assisted research programme; not peer reviewed.

Abstract

We study the law $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ ($\kappa = \frac{1}{2}$, fitted) with the kernel $\nu(y) = 1/(1 - e^{-\sqrt{y}})$ and a cold fluid of fixed cosmic amount. In particle-mesh simulations, letting the law act wherever matter has collapsed makes small-scale structure too clumpy: the excess in the matter power spectrum at $k \leq 1 \, h \, \text{Mpc}^{-1}$ is $+15.3\%/+19.3\%$ on the two footings at 256^3 and $+17.4\%$ at 512^3 , against an allowed 10%. We test one rule with no tuned radius. The law’s extra (phantom) mass is the cold fluid itself, so a halo can build the phantom only until its own cold fluid is used up; for an undepleted halo this happens at $r_{\text{edge}} = r_M / \ln[1/(1 - f_b)] \approx 5.85 \, r_M$. The cold fluid used is drawn from the same halo’s turnaround sphere, so mass is conserved halo by halo. With this rule the excess is 2.7% at 256^3 and 3.3% at 512^3 , with σ_8 within 0.6% of the Newtonian control. It holds in three random realisations, on both footings, with a_0 tracking dark energy, and when the two settings inherited from earlier work are varied. Turning the mass conservation off restores the excess (15.4%). Earlier rules that set the edge by hand, or from measured baryon censuses, sit at the 10% limit at 512^3 . The rule’s only inputs are κ and the cosmic baryon fraction. The cold fluid is handled as bookkeeping, not as simulated particles, and κ , the cold fluid’s amount and ρ_Λ remain undetermined.

1 Setting

The acceleration scale is $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ with $\kappa = \frac{1}{2}$ fitted. We report two footings, 9.3603×10^{-11} and $1.1312 \times 10^{-10} \, \text{m s}^{-2}$, separately. The phantom density is $\rho_{\text{ph}} = \nabla \cdot [(\nu - 1)g_b]/4\pi G$. A cold fluid with the cosmic amount $\Omega_c/\Omega_b = 5.364$ is assumed; no particle species is specified and the amount is an input. In the working model the phantom is not extra mass from nowhere: it is cold fluid that has settled into the profile the law requires.

2 The problem

Particle-mesh runs: box $200 \, h^{-1} \, \text{Mpc}$, initial redshift 49, 150 steps, the same initial conditions as a Newtonian control. Pass means $|\sigma_8 - 1| \leq 5\%$ and $\max |P - 1| \leq 10\%$ for $k \leq 1 \, h \, \text{Mpc}^{-1}$, both relative to the control.

rule	σ_8 ratio	$\max P - 1 $
phantom added as extra gravity (256^3)	1.2049 / 1.2573	
reservoir rule with gas filter (256^3)	1.0193 / 1.0243	0.153 / 0.193
the same at 512^3 (canonical)	1.0226	0.174

The excess grows with resolution. In the 256^3 snapshots 0.805 of the switched-on mass in filament-like cells (about 81%) lies inside resolved halos’ turnaround spheres, so the excess is a halo-outskirt effect. Galaxy lensing (KiDS-1000 isolated lenses, with a free two-halo amplitude) only requires the law out to $x = 0.48$ (canonical) and 0.53 (alternative) of the turnaround radius, and in our snapshots 0.83 of the excess source lies beyond $x = 0.4$.

3 The rule

For a point mass with this kernel the enclosed phantom is $M_{\text{ph}}(< r) = M_b/(e^s - 1)$ with $s = r_M/r$ and $r_M = \sqrt{GM_b/a_0}$. If the phantom is the halo’s own cold fluid, it can grow only until it equals that cold mass. For a halo that has kept its cosmic share of baryons this gives

$$r_{\text{edge}} = \frac{r_M}{\ln[1/(1 - f_b)]} = 5.85 \, r_M,$$

and the general form with retained fraction f_{ret} is $r_M / \ln(1 + f_{\text{ret}} f_b / (1 - f_b))$ (equal to $5.850 r_M$ for $f_{\text{ret}} = 1$). Version 1.0 used measured retention ($f_{\text{ret}} = 0.07\text{--}0.10$), which places the edge at 0.24–0.54 of the turnaround radius for KiDS lenses and 0.24–0.33 for the simulated groups and clusters. The simulation does not remove baryons, so the self-consistent choice there is $f_{\text{ret}} = 1$.

In the simulation, inside each halo’s edge the excess of the phantom over the local cold density is added as a source. The same amount is then removed from the cold fluid inside that halo’s turnaround sphere (in proportion to where the cold fluid is), so the added source sums to zero on every catchment. Nothing in this rule is tuned: there is no confinement radius and no catchment width. Two settings are inherited from earlier lanes: the width of the collapse switch and a gas-phase filter on the baryon source. We vary both below.

4 Results

run	σ_8 ratio	$\max P - 1 $
256 ³ , canonical / alternative	1.0033 / 1.0031	0.027 / 0.029
256 ³ , two more realisations (canonical)	1.0024 / 1.0026	0.022 / 0.017
256 ³ , the same two (alternative)	1.0023 / 1.0030	0.025 / 0.023
256 ³ , a_0 tracking dark energy	1.0035 / 1.0032	0.029 / 0.030
256 ³ , switch width halved / doubled	1.0033 / 1.0033	0.027 / 0.027
256 ³ , other gas filters	1.0027 / 1.0038	0.026 / 0.031
512³, canonical	1.0045	0.033
512 ³ , alternative	1.0054	0.040
512 ³ , a_0 tracking dark energy	1.0045	0.033
control: mass conservation switched off (256 ³)	1.0326	0.154
edge set by hand at $0.4 r_{\text{ta}}$ (512 ³)	1.0046 / 1.0059	0.080 / 0.0996
edge from census retention (512 ³)	1.0072 / 1.0073	0.092 / 0.1009

Every run of the rule passes, 15 of 15. The excess changes from 2.7% to 3.3% between 256³ and 512³, so it does not grow toward the limit as the earlier rules did. No catchment is ever asked for more cold fluid than it holds. The largest share of a catchment’s cold fluid used is 30% at 256³ and 57% at 512³; that share rises with resolution and is the quantity to watch at higher resolution. The control shows that the mass conservation, not the edge alone, removes the excess.

5 What this does and does not show

It shows that within this framework, structure growth can be made consistent with the Newtonian benchmark by a rule that follows from treating the phantom as conserved cold fluid, with no radius or width chosen to fit. The inputs are κ (fitted to rotation curves) and f_b .

It does not show the following:

- The cold fluid is bookkeeping in these runs. Its settling into the phantom profile is imposed, not derived from an action.
- One random realisation was run at 512³; the mesh does not resolve galaxy scales.
- The pass criteria compare with a Newtonian control. They are not a comparison with survey data.
- κ , the cold fluid’s amount and ρ_Λ are not derived, and no dark-matter particle is identified.

6 Reproducibility

Each test’s pass rule was committed before the run, and each test includes a deliberately broken control. The lanes are CFG414, CFG416 and CFG422–CFG427, CFG425 and CFG439 in the repository’s `campaign_fresh_gravity` directory; the engine is `CFG424_turnaround_catchment/cfg424_pm.py` (mode `RC = 0`).